INTERNAL SIGNALS AS JUDGES OF HALLUCINATED TOOL CALLS

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ABSTRACT

When a language model agent calls the wrong tool, the call is usually well formed, its arguments are of the right type, and its token probabilities are unremarkable, so the runtime executes it and an action is taken before anyone can review it. A guardrail must therefore judge the call at the moment it is generated, using only what the model itself exposes in a single forward pass. We study how well the model's own internal signals serve as that judge. We compare the published detectors of both internal families, probes on the residual stream and diagnostics of the attention graph, together with several related detectors, on identical data, splits and labels across a range of small models, attention designs and tool-calling benchmarks, and we evaluate every detector against a panel of deliberately uninformative baselines. Three results emerge. Whether an internal judge is needed at all depends on the model family rather than on its size: on some families output confidence is uninformative and a trained internal judge is the only thing that works, while on others confidence already tracks the model's errors and no internal judge we trained improves on it. Where internals are needed, probes on the residual stream at the structural positions of the call are the strongest judge we measured, and diagnostics of the attention graph come within a few points of them, but only when the attention heads are analysed separately. We explain why the head averaging that most of this literature performs is the limiting step: algebraic connectivity is a concave function of the attention graph, so averaging over heads can only overstate how well connected the graph is, with equality precisely when every head shares a Fiedler vector, and we measure how far from that condition real layers sit. Finally, although the two families carry partly independent evidence, no combination of their scores that we tried improves on the better one, and we report that negative result together with its operating points. The outcome is a practical rule stated in terms of the model to be deployed and what its serving stack can expose, rather than a ranking of methods.

1 INTRODUCTION

Language model agents act by calling tools. A wrong call is rarely obvious at the moment it is made: it parses, its arguments have the right types, it satisfies the schema, and the probabilities the model assigned to its tokens look ordinary. The runtime has no reason to hesitate, so a transfer is made or a resource is provisioned before a person sees it. Anything that catches such an error must therefore act in the interval between the model producing the call and the system executing it. That interval is too short to sample several continuations, to consult a second model, or to check an external source, and the only evidence available in it is the state of the model that produced the call.

This paper asks how good a judge that internal state can be. The question has a practical form as well as a scientific one, because different signals require different access. Serving stacks expose token probabilities almost everywhere, attention weights with some effort, and residual-stream activations only rarely. A detector that needs activations and one that needs only attention are therefore not alternatives competing for the same place in a system; they belong to different deployments. We accordingly report our results as a frontier over levels of access rather than as a ranking of methods.

Two lines of work have proposed internal judges for this task. One trains a probe on the residual stream at the structural positions of the generated call, the function name, the argument values, and the closing delimiter. The other reads a layer of attention as a weighted graph and computes spectral properties of it. Both report strong results, both need only a single forward pass, and because they consume different tensors neither has served as a baseline for the other. Several related detectors read hidden-state covariance, cross-layer dynamics of the residual stream, per-token activations with token-level supervision, or the balance of attention between the prompt and the generated text. This paper places them on the same data, splits and labels, and measures each against baselines built from surface properties that carry no information about the model's internal state.

Three findings organise the results. The first concerns when internal signals are worth reading at all. On the model family that hallucinates most, averaged token log-probability is useless and falls to chance or below, while probes on internal state are strong: these are models that make wrong calls confidently, which is the case a guardrail exists to catch. On the more recent families we tested, the picture reverses. Those models fail far less often, they express appropriate uncertainty when they do, and averaged log-probability then outscores every trained detector we measured. The reversal is not a curiosity, because it decides whether an operator should build an internal judge at all, so we test the obvious explanation for it. A trained probe needs labelled failures and confidence needs none, so a family that rarely fails supplies few positives to train on. We measure the relationship between the number of labelled failures and the advantage of a probe over confidence across runs, and we thin the positives of a well-supplied run to see whether scarcity alone reproduces the reversal. It does not: the probe survives a tenfold reduction of its labels, so the split is a property of the model family and not of the label budget, and we say plainly that with five families we cannot yet name the property.

The second finding is that among trained detectors, probes on the residual stream at the structural positions of the call are the strongest judge we measured, and that diagnostics of the attention graph come within a few points of them. An operator without access to activations is therefore not restricted to reading confidence, which is the practically useful half of the comparison. Both findings hold when the call is made with a conversation behind it: on the multi-turn categories of the benchmark, replayed with the correct earlier calls or with one deliberately wrong one in the history, the judges score as they do on a fresh prompt, and a judge that has never seen a corrupted history is not fooled by one.

The third concerns how attention must be read. Nearly all attention-graph diagnostics average the attention matrix over heads before computing anything, and that step, rather than the choice of spectral quantity computed afterwards, determines whether the diagnostic works at all. Analysing the heads separately moves the attention family from below a surface baseline to within reach of the residual stream, and a ladder of controls that hold the number of features fixed while varying the resolution shows that it is resolution, not dimensionality, that does the work. Section 4 explains why this must be so. The graph Laplacian is linear in the adjacency matrix and algebraic connectivity is a pointwise minimum of linear functionals of it, hence concave, so averaging over heads reports a graph at least as well connected as the typical head, with equality exactly when all heads share a Fiedler vector. We measure how far from that condition real layers sit, and find the averaged graph reports connectivity well above the mean over heads in every layer we examined. A companion result requires no assumptions at all: any function of the averaged graph is constant across head configurations with the same mean, so the loss belongs to averaging rather than to any particular readout.

The same analysis accounts for a baseline we do not surpass. The strongest published attention-only detector survives our stricter protocol and remains within noise of ours. Its authors note that the causal Laplacian is triangular, so its eigenvalues are its diagonal; we draw out what follows, namely that the features are a per-position degree sequence, invariant to any rewiring of attention that preserves in-degrees, whereas the symmetrised spectra we read are global functionals of the routing. That two such different quantities perform alike once heads are kept apart is the clearest evidence that head resolution, not the readout, carries the signal.

Finally we examine whether the two families should be combined, which is the natural question once both are shown to work. They are demonstrably not redundant: each retains substantial predictive signal after the other is regressed away. Yet nothing we tried converts that independence into a better detector. Concatenating their features does not help, stacking their scores does not help, and requiring both judges to agree does not help either once it is compared against a single judge at the recall the rule actually achieves rather than at a higher one. We report this as a negative result with its operating points rather than explain it away, and the recommendation that follows is to choose the judge the access tier allows and spend the labelling effort on its threshold.

Contributions.

1. A controlled comparison of internal judges for hallucinated tool calls, covering the published detectors of both families and several related ones, across small models spanning several attention designs and two tool-calling benchmarks in four configurations including multi-turn conversations with a corrupted-history condition, under splits that hold out whole tools and against a panel of uninformative baselines.
2. The finding that whether a trained internal judge beats untrained output confidence divides cleanly by model family and is not explained by the number of labelled failures, by size, or by attention design, tested across runs and within one run by thinning its positives; the practical consequence is a measurement to make on the model to be deployed before choosing a guardrail.
3. An account of why head averaging limits attention-graph diagnostics: an annihilation result that holds for any averaged readout, a concavity result that fixes the direction of the error and its equality condition, a measurement of how far real layers sit from that condition, and a ladder of controls showing that the gain from keeping heads apart is a matter of resolution and not of feature count. We also draw out, from its authors' own observation that the causal Laplacian is triangular, that the strongest published attention-only detector reads a degree sequence rather than a spectrum, which explains why two very different readouts perform alike.
4. An analysis of whether the two families combine, showing that they are far from interchangeable and yet that feature concatenation, score stacking and an agreement rule all fail to improve on the better single judge once compared at matched recall, together with the observation that precision is bounded by the model's failure rate.
5. An audit of the measurement chain itself, listing the defects that make results in this area fragile together with the guards that catch them.

2 Related work

Hallucination in tool use. Tool-calling benchmarks measure whether a model can call the right function with the right arguments (Patil et al., 2024; Qin et al., 2024; Patil et al., 2025), and the current leaderboard adds an explicit category in which no available tool applies and the model is expected to abstain (Patil et al., 2025). These evaluate capability. The question here is different: given that a model sometimes fails, can the failure be detected from the model's own state before the call executes. Healy et al. (2026) first showed that it can, with a probe on residual-stream states at the function-name, argument and delimiter positions, and that probe is the residual-stream baseline we compare against and extend. Concurrently, Yeats et al. (2026) train linear probes on the hidden states of eighteen models on the same benchmark and find that the probes catch argument errors of the right type and wrong value and generalise to unseen error kinds. That work establishes breadth on the residual tier; it has no attention tier, no surface floor and no held-out-tool split, which are the three things that decide the comparison this paper makes.

Probing the residual stream. Supervised probes on hidden states detect factual error (Azaria & Mitchell, 2023; Marks & Tegmark, 2024), predict hallucination risk from the query alone (Ji et al., 2024), and stand in for sampling-based estimates at the cost of a single pass (Kossen et al., 2024). Orgad et al. (2025) showed that the signal concentrates at the exact-answer tokens rather than being spread over the response, which is the observation the token-role probe applies to a call's name, arguments and delimiter. The family has been developed in several further directions: covariance and eigenvalue statistics of the states (Chen et al., 2024), token-level probes trained on entity-level annotations that operate in real time on long-form generation (Obeso et al., 2025), cross-layer dynamics of the residual stream (Zhang et al., 2025), spectral analysis of hidden-layer trajectories (Li et al., 2025), and probes over factual claims (Bhatnagar et al., 2026). Process reward models verify reasoning steps with a similar supervised signal (Lightman et al., 2024). All of these require residual-stream access. Our contribution to this family is not a new probe but the observation that the token-role probe remains the strongest single detector we measured, and that the correct comparison for it is an attention-only method at matched protocol.

Reading attention. Attention is the tensor practitioners are most likely to be able to read, and it has been used to decide cache eviction (Zhang et al., 2023), to detect hallucination through the ratio of attention paid to context against generation (Chuang et al., 2024), to aggregate multi-layer attention into a single map (Abnar & Zuidema, 2020), and to score consistency of attention across depth (Badash et al., 2026). The spectral line treats a layer as a weighted graph or kernel: Sriramanan et al. (2024) read eigenvalues of attention kernel maps and of hidden-state covariance in a single pass, Anonymous (2025) introduced Laplacian diagnostics of the attention graph, and Binkowski et al. (2025) obtained the strongest attention-only results to date from top-k Laplacian values; the same group has since read hallucination from attention sinks (Binkowski et al., 2026), a per-head signal we do not evaluate here. Dahlem et al. (2026) prove that every transpose-invariant spectral diagnostic of attention is blind to the direction of attention flow and evaluate spectral detectors under length control; every spectrum in this paper is of a symmetrised graph and is therefore orientation-blind in their sense, which bounds what the attention tier can see and is why our length floor sits in every table. Two features of this literature motivate our theory section. First, nearly all of it averages over heads before computing anything, which Section 4 shows is the decisive step. Second, part of it computes quantities that require residual states as a graph signal while presenting them as properties of attention; we separate the two and tag every feature by the tensors it consumes.

Sampling and output-side signals. Semantic entropy and self-consistency detect hallucination by sampling several continuations (Farquhar et al., 2024; Manakul et al., 2023). They are strong but they multiply inference cost and they require the freedom to sample, which an agent about to execute an action does not have. We therefore restrict this study to a single forward pass and include the cheapest output-side signal, mean token log-probability, as a baseline rather than as a competitor.

Position of this work. The two internals families were developed independently, and the open question was which to deploy. We answer it as a frontier over access rather than a ranking, and in the course of answering it we find that the interesting axis is not residual versus attention but averaged versus per head. The negative half of the result is as informative as the positive half: at matched protocol our per-head detector does not beat Binkowski et al. (2025), and Section 4.4 says why the two should be close despite computing different functionals.

3 Method

3.1 Task

A model is given a user request and a set of tool schemas in its own chat format, and generates greedily. The generated span either contains a tool call or does not. We label the call by comparing it against the benchmark's ground truth after normalising both sides to a canonical name and argument dictionary, and we record a *failure mode* rather than a bit: `valid`, `wrong_name`, `wrong_arg_values`, `missing_args`, `missing_calls`, `no_call` (prose instead of a call), `unparseable_call`, and, on the benchmark category where no tool applies, `valid_nocall` and `over_trigger`. The binary label is `valid` and `valid_nocall` against the rest.

The modes are not bookkeeping. Two of them, `no_call` and `unparseable_call`, are format failures that a length feature can detect without any access to internals, so every result is reported both on the full set and on the *semantic* subset that excludes them, where the detector must separate a correct call from a well-formed but wrong one. On the benchmark that mixes a category with no applicable tool, we report a third column restricted to the *call-expected* items, because on the remaining items the label reduces to whether a call was emitted at all.

3.2 Access tiers

Deployments differ in what they can read from a model, and the difference is not cosmetic. Token log-probabilities are returned by every serving API. Residual-stream states are a forward hook away in any stack that runs the model's own code, but are not returned by hosted APIs. Attention probabilities are the most awkward of the three: fused attention kernels never materialise them, so reading them requires the eager attention path or a second pass, which is what we do (Appendix C). The attention tier is therefore not the cheapest tier to reach; its interest is that it needs no activations, so a judge built on it can be trained and audited from attention maps alone, and that its features have a graph-theoretic reading that the theory of Section 4 can address. We group detectors by what they consume and report a frontier over tiers rather than a single ranking, and we measure what each costs at the moment the call is produced (Appendix F).

Logits. Mean token log-probability of the generated span, with the sign fixed on training folds only.

Attention. Functions of the attention weight matrices alone (Section 3.3).

Residual. Functions of the residual-stream states, that is, the block outputs that `output_hidden_states` returns (Section 3.4).

We keep the tiers strictly separate, which requires care. A family of published "attention" diagnostics projects residual-stream vectors onto the attention eigenbasis, giving Dirichlet energy, smoothness, spectral entropy and high-frequency energy ratio (Anonymous, 2025). Those quantities require both tensors and cannot be computed under attention-only access. In prior work built on that library, four of five nominally spectral features were of this hybrid kind, which inflates an attention-only claim with residual-stream information. Appendix B tags every feature by the tensors it consumes.

3.3 Attention-only detectors

All of the following are computed on the subgraph induced on the generated call, using Eq. 1 and the normalised Laplacian.

Per-head spectral profile (ours). For each layer ℓ and head h we take one eigendecomposition of $\mathcal{L}_{\ell h}$ and read five eigenvalue-only quantities: the Fiedler value λ_2, the connectivity ratio $\lambda_2/\lambda_{\max}$, the normalised spectral entropy of the eigenvalue distribution, the fraction of eigenvalue mass in the upper half of the spectrum, and the spectral radius $\lambda_{\max}$. Because all five come from the same decomposition, five cost what one costs. The feature vector is the $L \times H \times 5$ array. We also evaluate each of the five alone, to see which carries the signal.

Cross-layer dynamics (ours). Per-head differences between consecutive layers, $\phi_{\ell+1,h} - \phi_{\ell,h}$, for a chosen ϕ. A flat per-layer feature set represents depth only implicitly; the differences make the evolution explicit, and the theory in Section 4.4 predicts that a purely local feature family cannot express it.

Trajectory and velocity summaries. The head-averaged families that the prior literature uses, retained as ablations rather than as competitors: per-layer metric values, trajectory statistics over layers (mean, dispersion, slope, range, extremal steps), and layer-to-layer derivatives. Section 5.2 contrasts them with the per-head profile on identical data.

LapEigvals (Binkowski et al., 2025). The published attention-only state of the art, run from the authors' released implementation: per-layer, per-head, position-normalised in-degree minus self-loop, sorted, top-k kept, with k selected on validation from $\{5, 10, 25, 50, 100\}$ and a balanced logistic regression on the concatenation. Section 4.4 discusses what these features measure.

3.4 Residual-stream detectors

Token-role probe (Healy et al., 2026). The published state of the art for this task. At eight evenly spaced depths, the residual state at the function-name token, the mean over argument-value tokens, and the state at the closing delimiter are concatenated, and a classifier is trained on the concatenation. We evaluate both the original single-hidden-layer network and a regularised linear probe, and the token positions are located inside the generated span only.

Gram spectrum (Chen et al., 2024). Eigenvalues of the covariance of the residual states over the generated span, together with effective rank and a log-determinant surrogate, following the EigenScore family.

3.5 EVALUATION PROTOCOL

Grouped splits at the level of the tool. Examples are grouped by ground-truth tool name and split at the group level in a five-fold cross-fit, so every score is on tools absent from training. This is stricter than splitting by prompt, and it is necessary: with prompt-level splits, a classifier built from the tool identity alone reaches 0.323 AUC on one of our runs, because errors concentrate on particular tools. Under tool-level splits the same classifier is at or below chance everywhere.

Pooled cross-fit scoring. Each example is scored exactly once, by a model that never saw its tool, and we report one AUC pooled over the whole dataset rather than the mean of per-fold AUCs, which on this data would be computed on folds containing a handful of positives. We repeat the whole procedure over five split seeds and report the mean and standard deviation across seeds.

Selection discipline. Every hyperparameter, the direction of every unsupervised score, and the k of LapEigvals are chosen on training and validation folds only. The one place a sweep over features appears, we select the feature on training folds and report it on test, and label it as such.

Confound panel. Alongside the detectors we train three deliberately uninformative classifiers: prompt and generation lengths plus a truncation flag, generation length alone, and a one-hot encoding of the ground-truth tool. A detector is only interesting to the extent that it exceeds these.

Labels are recomputed at evaluation time. Labels are a function of the labelling code, not of the extraction run, so the evaluator re-derives every label from the stored generation. This is what let us discover, after the fact, that an earlier labelling rule counted a model's use of its own current date as a hallucination.

4 THEORY

Almost every attention-graph diagnostic begins by averaging the attention matrix over heads, and then reads a spectral quantity off the single graph that remains (Anonymous, 2025; Li et al., 2025; Badash et al., 2026). This section shows that the averaging step, not the choice of spectral quantity, is what decides whether the diagnostic can see a failure at all. Three statements follow. The first is about any functional of the averaged graph, the second gives the direction and the equality condition of the error that averaging makes for algebraic connectivity, and the third draws out what the features of the strongest published attention-only detector measure, which is a degree sequence rather than a global spectrum.

4.1 SETUP

Fix one layer with H heads and T token positions. Head h produces a row-stochastic attention matrix $A_h \in \mathbb{R}^{T\times T}$, which we symmetrise into a weighted adjacency and strip of self-loops,

$$W_h = \tfrac{1}{2}\left(A_h + A_h^\top\right) - \operatorname{diag}\!\left(\tfrac{1}{2}(A_h + A_h^\top)\right), \tag{1}$$

with combinatorial Laplacian $L_h = D_h - W_h$, where $D_h = \operatorname{diag}(W_h\mathbf{1})$, and symmetric normalised Laplacian $\mathcal{L}_h = I - D_h^{-1/2} W_h D_h^{-1/2}$ on the support of D_h. The *head-averaged* graph is $\bar{W} = \frac{1}{H}\sum_h W_h$, with $\bar{L}$ and $\bar{\mathcal{L}}$ built from it. We write $\lambda_2(\cdot)$ for the second-smallest eigenvalue, the algebraic connectivity, and $\lambda_{\max}$ for the largest. A *per-head* read of the layer keeps the vector $(\phi(W_1), \ldots, \phi(W_H))$ for a scalar functional ϕ; a *head-averaged* read keeps the single number $\phi(\bar{W})$.

4.2 AVERAGING IS A LOSSY PROJECTION, WHATEVER IS READ OFF IT

Proposition 1 (annihilation)**.** *Let Φ be any function of the head-averaged graph, that is, any map of the form $\{W_h\}_{h\le H} \mapsto \Phi(\bar{W})$. Then Φ is constant on every set of head configurations sharing the same mean. In particular, for every $H \ge 2$ and every $T \ge 3$ there exist configurations $\{W_h\}$ and $\{W'_h\}$ with $\bar{W} = \bar{W}'$ whose per-head connectivities differ by $\max_h \lambda_2(L_h) - \max_h \lambda_2(L'_h) = \Omega(1)$, so no head-averaged read can separate them while a per-head read separates them by a constant.*

Proof. The first claim is immediate, since Φ factors through $\bar{W}$. For the second, take $T \geq 3$, let P be the path graph on T nodes with weights $\frac{1}{2}$ and let K be the complete graph with weights $1/(T-1)$. Both have row sums at most one, so each is the symmetrised, self-loop-free adjacency (Eq. 1) of some row-stochastic attention matrix, and so is their mean. Put $W_1 = P$, $W_2 = K$ and $W_1' = W_2' = \frac{1}{2}(P{+}K)$, and for $H > 2$ repeat the pattern. Both configurations have mean $\frac{1}{2}(P{+}K)$. The path has $\lambda_2(L_P) = 1 - \cos(\pi/T) = \Theta(T^{-2})$ while the complete graph has $\lambda_2(L_K) = T/(T-1) > 1$, so the per-head vectors differ in their maximum by more than $1 - \Theta(T^{-2})$, whereas every head-averaged functional returns the same value on both. $\square$

Proposition 1 is elementary, and we state it because the literature it addresses has behaved as though it were false: averaging is a many-to-one map, and the proposition only makes precise what that map cannot preserve and shows the loss is of constant size on admissible graphs. It says nothing about which spectral quantity is read, so it applies to the Fiedler value, to the whole eigenvalue profile, to Dirichlet energy, and to any statistic of the averaged graph that a future paper might propose. The information that distinguishes the two configurations is carried by the *contrast between heads*, and the mean is precisely the statistic that discards contrast. Whether real failures are visible in that contrast is an empirical question, which Section 5.2 answers, but no averaged read can answer it either way.

4.3 The direction of the error, and when it vanishes

Proposition 1 exhibits a worst case. The next statement is about the typical case, and it fixes the sign of the discrepancy.

Proposition 2 (head averaging overstates connectivity)**.** *For any heads $W_1, \ldots, W_H$,*

$$\lambda_2(\bar{L}) \;\geq\; \frac{1}{H}\sum_{h=1}^{H} \lambda_2(L_h), \tag{2}$$

with equality if and only if the heads admit a common Fiedler vector, that is, a unit $x \perp \mathbf{1}$ attaining $\min_{y \perp \mathbf{1}, \|y\|=1} y^\top L_h y$ simultaneously for every h.

Proof. The Laplacian is linear in the adjacency, so $\bar{L} = \frac{1}{H}\sum_h L_h$. Since $L_h \mathbf{1} = 0$ and $L_h \succeq 0$, Courant–Fischer gives $\lambda_2(L) = \min_{x \perp \mathbf{1},\, \|x\|=1} x^\top L x$, a pointwise minimum of functions that are linear in L. A pointwise minimum of linear functions is concave, so λ_2 is concave on the convex cone of graph Laplacians, and Jensen's inequality applied to the uniform average over heads gives Eq. 2. For the equality case, let $x^\star$ attain the minimum for $\bar{L}$. Then $\lambda_2(\bar{L}) = \frac{1}{H}\sum_h (x^\star)^\top L_h x^\star \geq \frac{1}{H}\sum_h \lambda_2(L_h)$, and the inequality is tight exactly when $(x^\star)^\top L_h x^\star = \lambda_2(L_h)$ for every h, that is, when $x^\star$ is a Fiedler vector of every head. $\square$

The consequence is one-directional and therefore diagnostically bad in a specific way. A hallucinated call that fragments the routing of a few heads lowers λ_2 for those heads, but Eq. 2 bounds the averaged graph's connectivity *below* by the mean over heads, and the minimising direction of the average is a consensus direction that the fragmented heads need not share. Averaging thus reports a graph that is at least as well connected as the typical head, and the evidence of fragmentation is compressed toward the consensus.

The equality condition is a measure-zero event, so the question is not whether it holds but how far real layers sit from it, and that is measurable. On 210 layers drawn from 3 models and three inputs, the combinatorial inequality Eq. 2 holds with 0 violations, as it must, and the gap is bounded away from zero everywhere, with minimum 0.0026 and mean 0.057 (Figure 1). The mean pairwise absolute alignment between per-head Fiedler vectors is 0.45, far from the value 1 that equality requires, and the standard deviation of λ_2 across the heads of a layer averages 0.154. For the normalised Laplacian actually used by our detectors the Rayleigh quotient is not linear in W, so Eq. 2 is not a theorem; empirically it holds on all but 3 of the 210 layers, and the averaged graph reports connectivity a median 14% above the mean over heads, reaching $1.72\times$ at the extreme. Head averaging is therefore not a mild smoothing of a per-head quantity; it reports a systematically better connected graph than any head in the layer, and it does so by an amount that varies from layer to layer.

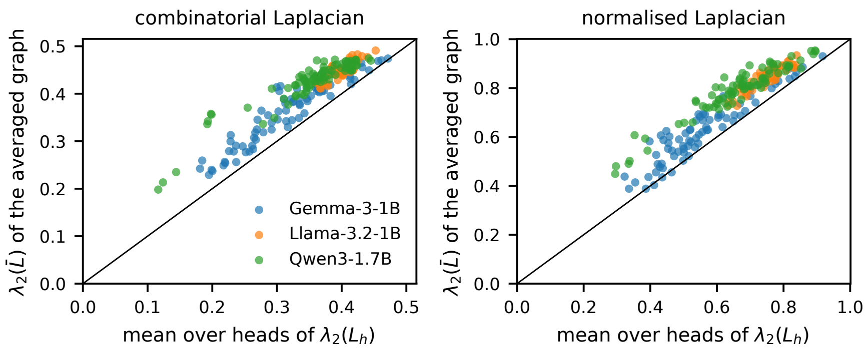

Figure 1: Proposition 2 measured. Each point is one layer of one model on one input; the horizontal axis is the mean over heads of the per-head algebraic connectivity and the vertical axis the connectivity of the head-averaged graph. Every point lies above the diagonal for the combinatorial Laplacian, as the proposition requires, and all but 3 do for the normalised Laplacian the detectors use, where the inequality is not a theorem. The distance from the diagonal is the connectivity that head averaging invents.

4.4 The strongest attention-only baseline reads degrees, not a spectrum

LapEigvals (Binkowski et al., 2025) is the strongest published attention-only hallucination detector. Its features are the top-k eigenvalues of the Laplacian $L = D - A$ of the unsymmetrised causal attention matrix, and its authors observe (their Lemma 1) that because $A_{ij} = 0$ for $j > i$ the matrix L is lower triangular, so its eigenvalues are its diagonal entries and no eigendecomposition is needed:

$$\mathrm{spec}(L) = \{\, d_j - A_{jj} \,\}_{j \le T}, \tag{3}$$

where d is their position-normalised in-degree. We take that observation as given and draw out what it implies for the comparison in this paper.

Remark 1 (LapEigvals is a degree statistic)**.** *By Eq. 3 the LapEigvals features depend on A only through the column sums and the diagonal. Any two causal attention matrices with the same in-degree sequence and the same self-attention weights have identical features, however differently they route attention between positions. The features are therefore a per-position, per-head degree statistic: local, invariant to in-degree-preserving rewiring, and carrying no eigenvector and no connectivity information. The symmetrisation in Eq. 1 destroys triangularity, so the spectra our detectors read are global functionals of the routing that change under such rewiring.*

This is not a criticism of the method, which works, and works well enough that our detectors do not beat it (Section 5.4). It is a statement about what the two families measure: they are not two estimates of the same object, and the fact that both keep heads apart is the one thing they share.

Corollary 1 (decorrelation prediction)**.** *If the two families measure different functionals, their scores should not be interchangeable: the rank correlation between them should be well below one, each should retain predictive signal once the other is conditioned on, and a combination of the two should exceed both. Section 5.5 tests the three predictions; two hold and the third fails.*

4.5 Why the call-local subgraph is length-invariant

The detectors read the subgraph induced on the generated call rather than on the whole sequence, which matters for a confound. Lengths differ systematically between correct and hallucinated calls, so any feature that scales with T can carry label information that has nothing to do with routing. The normalised Laplacian removes the crudest version of this.

Lemma 1 (bounded spectrum)**.** *For any weighted graph with non-negative weights and no isolated vertices, $\mathrm{spec}(\mathcal{L}) \subset [0, 2]$, with $\lambda_{\max} = 2$ if and only if a connected component is bipartite. The bound is independent of the number of nodes.*

Table 1: The runs. *Fail* is the fraction of generations that fail and *Pos.* the number of labelled failures. The attention column gives the number of layers, the number of query heads, and whether every layer computes softmax attention or only a subset does.

Model	Data	N	Fail	Pos.	Neg.	Attention
Llama-3.2-1B	Glaive	651	47%	303	348	16L, 32H dense
Llama-3.2-3B	Glaive	651	18%	119	532	28L, 24H dense
Gemma-3-1B	Glaive	651	29%	190	461	26L, 4H sliding
Llama-3.2-1B	BFCL	850	72%	612	238	16L, 32H dense
Llama-3.2-3B	BFCL	850	57%	488	362	28L, 24H dense
Qwen3-1.7B	BFCL	850	22%	186	664	28L, 16H dense
MiniCPM5-2B	BFCL	850	11%	92	758	42L, 16H dense
Qwen3.5-0.8B	BFCL	850	39%	328	522	24L, 6 attentive
Qwen3.5-4B	BFCL	850	14%	116	734	32L, 8 attentive
Qwen3.5-2B†	Glaive	740	3%	21	719	24L, 6 attentive
Llama-3.2-1B	BFCL-live	765	73%	559	206	16L, 32H dense
MiniCPM5-2B	BFCL-live	779	23%	178	601	42L, 16H dense

Proof. Standard (Chung, 1997): for $x \neq 0$, $\frac{x^\top \mathcal{L} x}{x^\top x}$ equals a ratio whose numerator $\sum_{i<j} W_{ij}(y_i - y_j)^2$ with $y = D^{-1/2}x$ is bounded by $2\sum_{i<j} W_{ij}(y_i^2 + y_j^2) = 2x^\top x$, with equality only in the bipartite case. □

Every per-head feature we use is a function of a spectrum contained in $[0, 2]$ whatever the length of the call, whereas the combinatorial spectrum grows with degree and hence with T. Length invariance by construction does not prove the absence of a length confound, since the routing itself may correlate with length, so Section 5.8 measures a surface-feature baseline built from lengths alone and reports every detector against it.

5 Experiments

5.1 Setup

Table 1 lists the runs. They cover five model families between 0.8B and 4B parameters and three data distributions: the Glaive function-calling corpus (GlaiveAI, 2024), the curated single-turn categories of BFCL-v4 (Patil et al., 2025) including the category where no tool applies, and the live categories of the same benchmark, which are queries contributed by its users rather than written for it. Three attention designs are represented: dense grouped-query attention at several head counts, a hybrid stack in which only a quarter of the layers compute softmax attention at all, and a sliding-window stack that attends locally in all but four of its layers and carries four query heads. Failure rates span an order of magnitude, from 2.8% to 71.8%, which lets us separate claims about detectors from claims about how often the models fail. The Qwen3.5-2B run yields too few labelled failures to support a conclusion and is reported only as a boundary case.

Each run is scored on the hardest population it supports. Where the benchmark contains the category with no applicable tool, that is the call-expected population, because on the remaining items the label reduces to whether a call was emitted at all; otherwise it is every parseable call. Calls that fail to parse because our generation budget cut them off are excluded throughout, as they measure our protocol rather than the model.

5.2 Head resolution

Table 2 is the paper's central measurement, and it isolates one variable. Every row reads the same attention tensors of the same generations under the same splits and the same seeds; the blocks differ only in whether the heads are averaged before the graph is built. The head-averaged column sits between 0.517 and 0.735 and on several runs does not clear the surface-length confound. The per-head block reaches 0.933 on Llama-3B and 0.904 on the same model on BFCL. Averaged over the

Table 2: One detector family per column, scored on the hardest population each run supports: the items where a call is genuinely required where the benchmark has an abstention category, and every parseable call otherwise. Best per row in bold. *avg. spectral* averages attention over heads before computing anything, as the prior literature does; *per-head* keeps the heads apart. *surface* is a classifier on lengths alone and is not a detector but a floor. Bold marks every detector within one seed standard deviation of the row's leader, since the ordering inside that band changes with the split seed; Appendix D gives the spread.

Model	Data	logprob	surface	avg. spectral	per-head	LapEigvals	Lookback	token-role	token-level
Llama-3.2-1B	Glaive	0.576	0.556	0.535	0.940	0.942	0.917	**0.957**	**0.963**
Llama-3.2-3B	Glaive	0.738	0.609	0.652	**0.930**	0.907	**0.911**	**0.937**	0.897
Gemma-3-1B	Glaive	0.647	0.699	0.716	0.856	0.854	0.765	**0.915**	**0.914**
Llama-3.2-1B	BFCL	0.623	0.754	0.773	0.873	0.873	0.861	**0.913**	0.903
Llama-3.2-3B	BFCL	0.692	0.632	0.757	0.910	0.895	0.892	**0.929**	**0.930**
Qwen3-1.7B	BFCL	0.537	0.599	0.795	0.892	0.905	0.911	**0.927**	0.908
MiniCPM5-2B	BFCL	**0.805**	0.546	0.635	0.730	0.722	0.771	0.788	0.724
Qwen3.5-0.8B	BFCL	**0.769**	0.616	0.599	0.657	0.627	0.634	0.708	0.674
Qwen3.5-4B	BFCL	**0.821**	0.637	0.620	0.665	0.726	0.692	0.749	0.737
Qwen3.5-2B[†]	Glaive	0.784	0.721	0.708	0.638	0.678	–	0.746	–
Llama-3.2-1B	BFCL-live	0.403	0.709	0.658	**0.816**	0.782	0.800	**0.808**	**0.808**
MiniCPM5-2B	BFCL-live	**0.776**	0.582	0.582	0.725	0.661	0.739	0.726	0.741

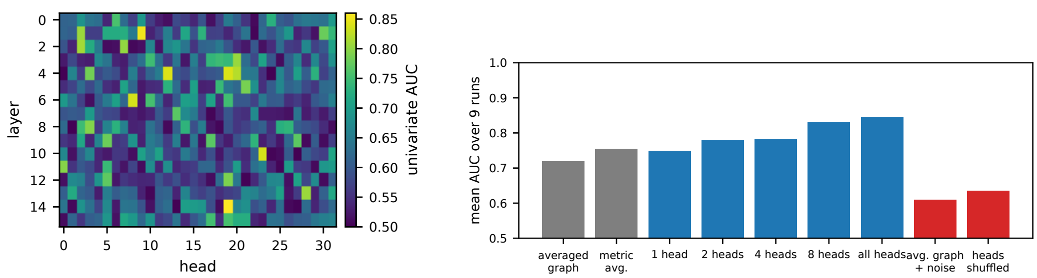

Figure 2: Left: where the attention signal lives. Univariate AUC of each head's spectral radius on the generated call, Llama-3.2-1B on Glaive, direction-free. The signal is concentrated in particular heads of particular layers; averaging over a row of this map mixes its best head with its worst. Right: the resolution ladder of Section 5.3, mean AUC over 9 runs. Grey bars are the two $L \times 5$ reads, blue bars keep heads apart with an increasing number of heads, and red bars have the full per-head width but no head information.

12 runs where both are available, keeping the heads apart is worth 13.4 AUC points, with a range of -7.0 to 40.5; the single negative entry is the hybrid model with too few positives to interpret.

This is what Section 4 predicts, and it is worth being clear about the direction of the inference. Proposition 1 says an averaged readout cannot in principle recover between-head contrast, and Proposition 2 says the averaged graph systematically overstates connectivity unless heads share a Fiedler vector, which they measurably do not. Neither statement guarantees that a failure is visible in the contrast. Table 2 establishes that it is.

5.3 Resolution or dimensionality?

The per-head profile has $L \times H \times 5$ features and the head-averaged block $L \times 5$, so the comparison in Table 2 varies two things at once. A ladder of controls separates them, all on the same stored features, the same regularised linear classifier and the same tool-grouped splits over 9 runs. *Metric averaging* computes every metric per head and then averages over heads, giving $L \times 5$ features like the literature's read but with the Jensen effect of Proposition 2 removed: it scores 0.755 against 0.719 for the averaged graph, a change of +0.036 that is positive on 6 of 9 runs. Keeping the heads apart adds a further +0.090 on top of that, to 0.845. Two controls hold the width fixed at the per-head value. Padding the averaged-graph features with Gaussian noise to the same width leaves them at 0.609, a change of -0.110 from the unpadded value, so width without information costs rather than buys. Permuting the head axis independently for every example, which keeps every value and

Table 3: The access frontier. One representative per tier, fixed before any result was seen: the mean log-probability, the five-metric per-head profile and LapEigvals for attention, the token-role probe for the residual stream, and the length classifier as the floor. Subsets as in Table 2.

Model	Data	floor	logits	Δ	attention: per-head	Δ	attention: LapEigvals	Δ	residual: t
Llama-3.2-1B	Glaive	0.556	0.576	+0.020	0.940	+0.384	0.942	+0.387	0.9
Llama-3.2-3B	Glaive	0.609	0.738	+0.129	0.930	+0.322	0.907	+0.298	0.9
Gemma-3-1B	Glaive	0.699	0.647	-0.052	0.856	+0.157	0.854	+0.155	0.9
Llama-3.2-1B	BFCL	0.754	0.623	-0.131	0.873	+0.119	0.873	+0.120	0.9
Llama-3.2-3B	BFCL	0.632	0.692	+0.059	0.910	+0.277	0.895	+0.263	0.9
Qwen3-1.7B	BFCL	0.599	0.537	-0.062	0.892	+0.294	0.905	+0.306	0.9
MiniCPM5-2B	BFCL	0.546	0.805	+0.259	0.730	+0.184	0.722	+0.176	0.7
Qwen3.5-0.8B	BFCL	0.616	0.769	+0.152	0.657	+0.040	0.627	+0.011	0.7
Qwen3.5-4B	BFCL	0.637	0.821	+0.184	0.665	+0.028	0.726	+0.089	0.7
Qwen3.5-2B†	Glaive	0.721	0.784	+0.064	0.638	-0.083	0.678	-0.042	0.7
Llama-3.2-1B	BFCL-live	0.709	0.403	-0.306	0.816	+0.107	0.782	+0.074	0.8
MiniCPM5-2B	BFCL-live	0.582	0.776	+0.194	0.725	+0.143	0.661	+0.078	0.7

destroys only which head it came from, drops the per-head profile to 0.635, a loss of +0.210. Finally a random subset of one, two, four and eight heads per layer scores 0.748, 0.780, 0.781 and 0.832, rising with the number of heads kept rather than jumping at the first. The signal is carried by which heads do what, and a read that cannot tell the heads apart cannot see it, whatever its width.

Which of the five per-head metrics carries the signal changes with the run: the spectral radius leads on Llama-1B with 0.940 against the Fiedler value's 0.892, and the ordering reverses on Llama-3B. Reporting the per-run best would be selection on test, so we recommend and report the five-metric profile, which is at or near the top in every run and costs the same single eigendecomposition.

5.4 The access frontier

Table 3 is the deployment result, and it has two regimes. On the Llama, Qwen3 and Gemma runs the ordering of the internal tiers is stable: the residual stream is best, attention-only trails it by a few points, and both are far above the confound floor, while the token log-probability reaches 0.693 on Llama-3B and 0.537 on Qwen3-1.7B, which is chance, on a run where the same model's residual stream supports 0.927. A guardrail built on output confidence would pass exactly the calls this paper is about on those models. On the MiniCPM5 and Qwen3.5 runs the picture inverts: the log-probability is the best single detector and no trained judge reaches it. Section 5.10 examines that division. Within the internal tiers the token-role probe leads the per-head profile on eleven of the twelve runs; on the one where attention leads, the difference is within seed noise. The weakness of confidence on the first group does not track model quality: Qwen3-1.7B abstains on the no-applicable-tool category almost as often as any model in the sweep (Section 5.7), and its confidence is the least informative we measured.

Within the attention tier, LapEigvals and our per-head profile trade places by run and stay within noise of one another. We do not claim to beat it. What the per-head profile offers instead is compactness, $L \times H \times 5$ features against $L \times H \times k$ with k up to 100, and a readout with a graph-theoretic meaning, which is what makes Section 4 possible. A smaller probe should also need fewer labelled failures, and Figure 3 tests that by thinning the positives available to the training folds on 6 runs while scoring on every example. The two attention judges are tied at full budget (0.906 against 0.896) and at the smallest budget, a handful of failures per fold (0.768 against 0.768); in between, with a tenth or a fifth of the labels, the per-head profile leads by +0.027 and +0.039, on 4 and 4 of 6 runs. The edge is modest and not uniform across runs, so we state it as parity with a small advantage in the low-label regime, not as a general label-efficiency result. The residual probe leads both throughout, 0.834 at a tenth of the labels against 0.811, and it loses less than either as labels shrink.

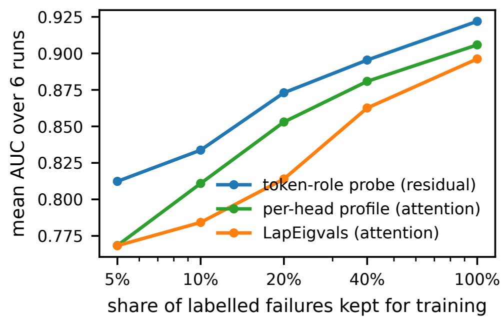

Figure 3: Label efficiency of the trained judges. Positives are thinned in the training folds only, every example is scored, and the curve is the mean over 6 runs of the Llama, Gemma and Qwen3 families. The two attention judges are tied at both ends and the per-head profile leads by a few points in between; the residual probe leads throughout.

5.5 Decorrelation of the two attention families

Corollary 1 makes three predictions from Remark 1. Two hold and one fails, and the failure carries the more useful lesson.

The first prediction, that the two families' scores are not interchangeable, holds. Over 11 runs the Spearman correlation between the per-head spectral score and the LapEigvals score averages 0.58, ranging from 0.23 to 0.87. They agree substantially, as two competent detectors of the same events must, but they are far from being the same function of the forward pass.

The second prediction, that each retains signal once the other is removed, holds strongly. Regressing the per-head score on the LapEigvals score and scoring the residual gives 0.693 AUC on average, and the reverse gives 0.634. Both are far above chance, so neither family is a noisy copy of the other: an in-degree sequence and a symmetrised spectrum carry substantially independent evidence about the same call, as Proposition 1 implies they should.

The third prediction, that a combination should therefore exceed both, fails. Concatenating the two feature sets changes AUC by +0.001 on average and improves on the better of the two in only 4 of 11 runs, with a best case of +0.047. For feature concatenation a variance account is at least available: the union carries thousands of features against a few hundred positives. The two-score combinations of Section 5.6 have no such excuse and fail too, so that account cannot be the whole story. The distinction matters for practice. Earlier work reported that combining internal signals does not help and read that as evidence that the signals are redundant. The residual analysis shows they are not redundant in the sense of being the same function of the forward pass. Whether they are redundant for the decision is a different question, and Section 5.6 answers it in the negative direction.

Because this is a negative result that happens to favour our own framing, the union is given the more generous of two protocols: its LapEigvals width is selected on validation rather than fixed. Fixing it instead gives -0.008, so the conclusion does not depend on that choice.

5.6 Combining the two judges

Section 5.5 shows that the residual-stream and attention judges carry substantially independent evidence, which makes it natural to ask whether running both is worth the cost. We tried three ways of combining them and none of them is.

Concatenating their features changes AUC by +0.001. Stacking their outputs with a combiner fitted on out-of-fold training scores, which is the standard remedy for a small validation fold and gives the combiner several times more labelled failures to work with, changes AUC by -0.022 and improves on the better single judge in 2 of 11 runs. Putting both scores on one scale first, by mapping each through the empirical distribution of its out-of-fold training scores, changes AUC by -0.011 and helps in 3 of 11 runs, with a best case of +0.015; every fitted combiner gave both judges positive weight. Part of the remaining deficit is the pooling penalty of Section 5.12, which a combination pays twice, so we also score the simplest combination inside each fold, where no such penalty exists: on Llama-1B on Glaive the average of the two judges' scores reaches 0.946 against 0.949 for the better single judge in the same folds, and on Llama-3B 0.961 against 0.969. The combination matches the better judge and does not exceed it.

The third is the one an operator would reach for without any fitting: set a threshold for each judge on training data and flag a call only when both fire. It appears to work, and the appearance is an artefact of the operating point. The rule fires on fewer calls than either judge alone, so it runs at lower recall, and lower recall buys precision for nothing. Compared against the stronger single judge *at the recall the rule actually achieves*, the agreement rule changes precision by -0.011 on average and helps in 2 of 11 runs. Moving one judge's threshold does the same work.

We therefore report a negative result without a mechanism we can defend. The two families are far from interchangeable, and yet nothing we tried converts that into a better detector, within folds or pooled. A shortage of labels is not the reason: the combiners here had hundreds of out-of-fold positives, and Section 5.10 shows a far larger probe surviving a tenfold thinning of its labels. The likelier reading is that the two judges disagree mostly on examples both already score correctly, and agree on the ones both get wrong. The recommendation that follows is to pick the judge the access tier allows and spend the labelling effort on its threshold rather than on a second judge.

Precision is bounded by how often the model fails. One property of these operating points matters more for deployment than any difference between detectors. Across our runs the failure rate ranges from 10% to 63%, and precision tracks it: on the models that fail most, a judge reaches precision in the high nineties at usable recall, and on the model that fails least, no judge we measured exceeds a fifth. A guardrail is therefore most deployable exactly where the model is worst, and on a well-behaved model the false positives it raises may cost more than the failures it catches. That is an argument for measuring the failure rate of the deployed model before choosing a guardrail, and it is the same measurement Section 5.10 already recommends for a different reason.

5.7 Architecture and abstention

Attention design changes what the attention tier can read, and the effect is smaller than the earlier literature suggests. On the sliding-window model, which has four heads and one key-value head and computes local attention in all but four of its layers, the per-head profile still reaches 0.880 against a confound floor of 0.716. A narrow, sparse attention stack therefore does not disable spectral detection. The hybrid families, in which only a quarter of the layers carry softmax attention, are where the tier weakens, and there the attention-only detectors fall to the neighbourhood of the residual probes rather than below the confound floor.

The clearest architectural effect we find is not about detectors but about the models. On the benchmark category where no available tool applies and the correct behaviour is to answer without calling anything, the two Llama models fabricate a call almost every time: 149 of 150 items at 1B and 150 at 3B, with scale making it worse rather than better. Every other family we tested abstains most of the time, and abstains at every size we tested it: Qwen3-1.7B on 132 items, Qwen3.5-4B on 138, MiniCPM5-2B on 130, and the smallest model in the study, Qwen3.5-0.8B, on 128. We inspected these decisions by hand: the Llama outputs are well-formed calls, and the others are prose answers to the question that was asked.

Abstention is therefore a property of the training recipe and not of scale, and an earlier version of this study, which had only the Llama family in it, drew the opposite and wrong conclusion that small models cannot abstain. The practical consequence is unchanged in form but narrower in scope: for a model family that does not abstain, the capability has to be supplied from outside the model, and a guardrail of the kind measured here is the only option. For a family that does abstain, the guardrail is guarding against a rarer event, which is exactly the regime that Section 5.10 shows makes a trained judge hard to fit.

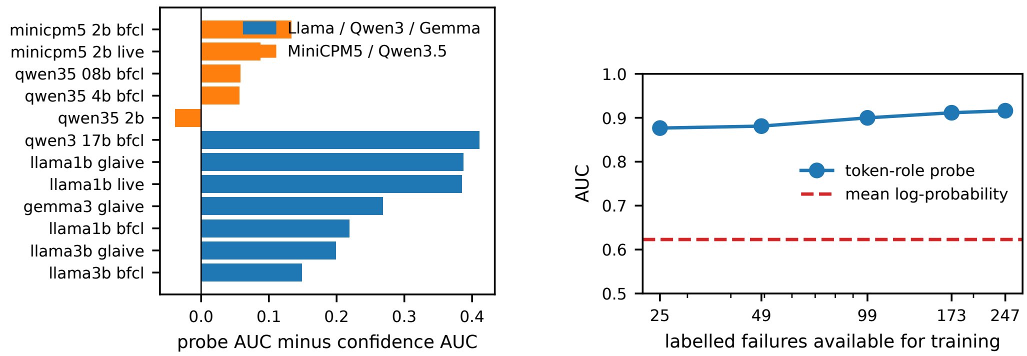

Figure 4: Left: the advantage of the best residual probe over mean log-probability on every run, scored on every parseable call. The two groups of families do not overlap. Right: the probe's AUC on Llama-3.2-1B on BFCL as its labelled failures are thinned, against the confidence baseline on the same data. A tenfold reduction does not bring the probe near the baseline, so the split on the left is not a shortage of labels.

5.8 Confounds and transfer

Every table reports the confound floor beside the detectors, and the floor is not always low: on Llama-1B on BFCL a classifier built from lengths alone reaches 0.835 on the full set, because format failures are long. That is why the semantic and call-expected subsets exist, and on them the floor falls to 0.733 while the detectors hold.

Transfer separates a detector that learned routing from one that learned a corpus. Trained on Glaive and applied unchanged to BFCL on the same model, the residual probe retains 0.904 on Llama-3B and the per-head profile 0.840, against in-domain values of 0.947 and 0.930. The length confound does not transfer at all: it falls to 0.469, below chance, on the same comparison. A feature that was carrying corpus-specific length regularities behaves exactly as one would expect under distribution shift, and the detectors do not.

5.9 Calls with a history

Every result so far concerns a single call made from a fresh prompt, and an agent does not work that way: its calls are made with a conversation behind them, and a wrong call can be inherited from an earlier one. We therefore evaluate on the multi-turn categories of BFCL-v4 under a teacher-forced protocol. A conversation is replayed with the ground-truth calls of the two preceding turns in the prompt, the model is asked for the calls of the current turn, and the label is whether those calls match the ground truth for that turn. This keeps the label unambiguous without running the benchmark's simulated back ends, at the price that the model never sees its own earlier mistakes. To restore that condition deliberately, half of the records with a history carry one preceding turn whose calls were replaced by a wrong call drawn from another conversation, presented as having succeeded, which we call the *cascade* condition. Two models are run, one from each side of the family split of Section 5.10: Llama-3.2-1B on 510 turns and MiniCPM5-2B on 538. Prompts run to several thousand tokens, so the attention features are reduced layer by layer inside a forward hook rather than materialised for the whole stack (Appendix C). Because a weak model echoes tool schemas into argument-less calls here, labels on these runs count arguments the schema does not define as a failure (Appendix A).

The models. Multi-turn calling is far harder for these models than the single-turn categories. MiniCPM5 fails on 54% of opening turns and 67% of all turns; Llama-1B fails on 97% of opening turns, 30% of its generations run past the budget, and the whole run leaves 35 well-formed correct calls, too few to train or score a judge on. We report Llama-1B as a boundary case, the multi-turn counterpart of the run in Table 1 that fails too rarely: a judge needs both kinds of example, and a model that almost never succeeds supplies no more of them than one that almost never fails. The cascade condition asks whether a wrong call in the history makes the next call worse, and on neither model does it. On turns with a history MiniCPM5 fails on 69% after a clean history and 70% after a corrupted one (Fisher $p = 0.760$ on 188 and 283 turns), and Llama-1B on 93% against 92% ($p = 0.853$). A single wrong call that the history reports as successful does not propagate; whether a wrong *result* would is a question this protocol cannot ask, and we return to it in Section 6.

The judges. Detection survives the history. On MiniCPM5, with tools held out as before, the token-role probe reaches 0.866, the per-head profile 0.777 and LapEigvals 0.764, against a length floor of 0.472; the probe scores turns with a history at 0.866 and opening turns at 0.865, so the history neither helps nor hurts it. The family split of Section 5.10 reappears intact: mean log-probability reaches 0.879 on this model and no trained judge exceeds it, with a history behind the call as without one. On Llama-1B every judge sits near chance on the 35 negatives available, LapEigvals highest at 0.755, and we draw no conclusion from that run.

The judge under cascade. The question a deployment cares about is whether a judge trained on well-behaved histories still works once the history contains a failure it has never seen. We train every judge on clean-history items only and score it, on held-out tools, separately on clean and on corrupted items. On MiniCPM5 the token-role probe scores 0.857 on clean and 0.859 on corrupted items, the per-head profile 0.728 against 0.751, and LapEigvals 0.737 against 0.718. A corrupted history does not change what a wrong call looks like from inside the model, at least not in a way that a judge trained without ever seeing one would miss; training on both conditions gives 0.864 on the corrupted items, the same figure. For an operator this is the useful half of a null result: the calibration set for a judge does not need to contain failed trajectories.

5.10 When a trained judge beats output confidence

Whether a trained internal judge is worth building at all is the first question an operator asks, and our runs do not give it one answer. They divide cleanly by model family (Figure 4). On the Llama, Qwen3 and Gemma runs a probe on the residual stream leads mean log-probability by a wide margin. On the MiniCPM5 and Qwen3.5 runs the margin nearly vanishes, and on the harder of our two evaluation subsets it reverses and confidence wins.

The division is complete rather than a tendency, and we are careful about the unit over which that is claimed. Over 12 runs scored on every parseable call, the probe's advantage averages +0.060 on the recent families against +0.288 on the others, and the two groups do not overlap: the largest advantage among the recent families, +0.133, falls below the smallest among the others, +0.149. Runs are not independent observations, however, since one checkpoint is evaluated on up to three benchmarks, so the unit for any test is the checkpoint, of which there are 8, 4 recent and 4 earlier. They too separate completely, which is the smallest attainable two-sided rank-test value at that size, $p = 0.029$. At the level of the family, which is what the claim is about, there are five observations and no test has power; we report the family means instead, Qwen3 +0.411, Gemma-3 +0.268 and Llama-3.2 +0.252 against MiniCPM5 +0.110 and Qwen3.5 +0.026, and let the reader judge the gap. Restricting to the items where a call is genuinely required, which removes the abstention decision and is the population this paper is about, the recent families average -0.046: confidence beats every trained detector we measured on 4 of 4 such runs. That subset has only 6 checkpoints, so its separation is descriptive. We report both subsets because the sign of the effect for the recent families depends on which is used, while the separation between families does not.

The obvious explanation is not the right one, and we report the test because we expected it to succeed. The recent families also fail far less often, and a supervised probe needs labelled failures while confidence needs none, so their probes might simply be starved. Two checks rule that out. First, thinning the labelled failures available to the Llama-1B run on BFCL from 247 to 25, a tenfold reduction, costs the probe only 0.039 AUC, from 0.916 to 0.877, which still leaves it far above the 0.623 that confidence reaches on the same data. A probe trained on twenty-five failures wins comfortably, so scarcity of labels is not what holds the other families back. Second, the apparent relationship between failures and advantage is an artefact of grouping. Across all runs the rank correlation between the number of labelled failures and the probe's advantage is 0.52 and does not reach significance at $p = 0.08$, and computed within each family separately it disappears, at +0.30

and -0.14. One run settles it alone: Qwen3.5-0.8B supplies 328 labelled failures, more than most runs where the probe wins, and confidence still leads there.

We therefore attribute the difference to the model family. The recent families we tested express uncertainty that tracks their own tool-calling errors, so their output distribution is already informative and a trained judge adds little; the older families do not, and there a trained judge is the only thing that works. With five families we cannot say which property of training produces this, and we do not claim a trend over time. Three confounds remain open and we name them rather than argue them away. The two recent families both emit their calls in an XML dialect while the others emit JSON, so the token probabilities being summarised are taken over different surface forms; a control that forces one family into the other's format would settle this and we have not run it. One of the two recent families has a hybrid attention stack, but the other, MiniCPM5, is a dense stack on which confidence wins just as clearly, so attention design is not the explanation. Finally, a model that has seen the benchmark in training will fail less and know when it fails, which is what we observe; the BFCL-live categories are contributed by users after the curated set and are the less likely to have been seen, and on them MiniCPM5 shows the same behaviour, which weakens the contamination account without excluding it. The practical consequence is nevertheless firm, and it is not the rule we set out to write: before building an internal guardrail, measure output confidence on a few hundred labelled calls from the model that will actually be deployed, because whether internals are needed is a property of that model and cannot be inferred from its size, its attention design, or its release date.

5.11 Does the comparison depend on how confidence is summarised?

Comparing a trained judge against "output confidence" presumes a way of summarising the output distribution, and the mean token log-probability is only one. If a better summary closed the gap on the families where confidence loses, the split of Section 5.10 would be an artefact of that choice rather than a property of the models. We therefore record 9 summaries per generated call, covering the mean, minimum, total and mean-lowest-five log-probability, perplexity, the fraction of low-probability tokens, and the mean, maximum and final predictive entropy, and score each as its own detector under the identical protocol with its direction fixed on training folds.

The choice matters, and it matters in the direction that weakens our own earlier framing. On Llama-3.2-1B on BFCL the mean log-probability reaches 0.626 while the best summary that is not confounded with length, the mean of the five least likely tokens, reaches 0.731. A single improbable token is what carries the evidence, and averaging over the call dilutes it. The probe's advantage therefore falls from +0.278 to +0.172. It does not disappear: the best available confidence summary still trails the probe's 0.903 by a wide margin. One summary must be set aside. The total log-probability of the span scores 0.745, which is close to the 0.753 of the length baseline on the same run, and since it grows with length it is largely measuring the same thing; we report it but do not treat it as a confidence result.

On the family where confidence wins, the choice of summary makes almost no difference. On MiniCPM5-2B the best summary reaches 0.808 against 0.805 for the mean log-probability, and the probe's deficit moves from -0.017 to -0.020. The summaries also agree with one another there, most of them falling within a couple of points, whereas on Llama they range over more than seventeen. That is a second and independent way of stating the split: on one family the model's expressions of uncertainty agree with each other and with its errors, and on the other they neither agree nor track the errors.

The consequence for the rest of the paper is a caveat rather than a correction. The cross-run comparison of Section 5.10 uses the mean log-probability, which we now know understates confidence on the families where it loses; the separation between families survives the fairer comparison, but it is smaller than a mean-log-probability comparison alone would suggest.

5.12 Statistical discipline

Pooled scores are conservative. Every AUC in this paper pools the out-of-fold scores of five classifiers, one per fold, each calibrated on its own training tools. A deployed judge is one classifier with one calibration, so pooling penalises a calibration drift between folds that a deployment never sees. We measured the cost on 3 runs by scoring the residual probe both ways: within folds it averages 0.932 on Llama-1B on Glaive against 0.905 pooled, and the cost averages 0.012 AUC across the runs. We keep the pooled protocol because per-fold AUCs on this data are computed on a handful of positives and are unstable, but the reader should take every number in Tables 2 and 3 as a floor on what a single calibrated judge achieves. The same effect is much larger for any score that is only meaningful relative to its fold, such as a rank: the within-fold rank average of the residual and attention judges scores 0.946 inside folds and 0.778 pooled, which is why Section 5.6 reads its combinations within folds.

Ties. Differences within the top group of detectors are smaller than the variation between split seeds. On Llama-1B a paired bootstrap over the pooled scores puts the residual probe, LapEigvals and the per-head profile within a three-way tie whose ordering changes with the seed. We therefore state the frontier between tiers, which is stable and large, and refrain from ranking detectors inside a tier, which is not.

6 DISCUSSION

Whether to build an internal judge is a question about the model. The result we did not expect is that the answer changes by model family, and changes completely (Section 5.10). On one group of families output confidence is uninformative and a trained probe is the only thing that works; on the other, confidence is already as good as anything we trained. The two groups separate without overlap on both of our evaluation populations, and the separation is not explained by how many labelled failures each model supplies, by its size, by its attention design, or by when it was released. We cannot say which property of training produces the difference, and with five families we would not try. What follows practically is a measurement rather than a rule of thumb: score output confidence on a few hundred labelled calls from the model that will be deployed, and build an internal judge only if that score is poor.

What the two internal tensors measure. Where internals do help, the residual stream at the structural positions of the call is the strongest signal, and it should be: those states are close to the decision the model is making, and a probe reads that decision almost directly. The attention graph is one step removed, describing how information was routed while the call was produced rather than what was concluded. That a routing summary recovers most of the probe's performance says the two are tightly coupled during a tool call, and the few points between them are a reasonable estimate of what the summary loses. The practical consequence is the frontier of Section 5.4: an operator who cannot reach the residual stream is not thereby reduced to reading confidence.

Resolution, not the readout. Two observations point the same way. Within the attention tier, the head-averaged readouts fail and the per-head readouts succeed on identical data (Table 2), controls that hold the number of features fixed attribute the difference to resolution rather than dimensionality (Section 5.3), and Section 4.4 shows that the strongest published member of that tier is an in-degree sequence rather than a spectrum, so the two best attention detectors agree on almost nothing except keeping heads apart. Within the residual tier the two detectors that read designated positions, the token-role probe and the token-level probe, are the strongest, while the two that pool over the whole span, a covariance spectrum and a cross-layer dynamics summary, are markedly weaker. In both tiers the detectors that preserve resolution win and the detectors that pool it away lose, which suggests the useful signal is local, carried by particular heads and particular positions, and that a practitioner's effort is better spent preserving that locality than selecting a statistic to compute on top of it.

Combining the two families. They are not redundant: each retains substantial predictive signal after the other is regressed away (Section 5.5). Yet nothing we tried converts that independence into a better detector. Concatenating features does not help, stacking scores does not help, and the rule an operator would reach for without any fitting, flagging a call when both judges fire, does not help either once it is compared against one judge at the recall the rule actually reaches. We do not have a mechanism for this that survives our own checks: a shortage of labels cannot be the reason, since a two-weight combiner had hundreds of out-of-fold positives on the runs where it failed and a far larger probe survives a tenfold thinning of its labels (Section 5.10). The two scores agree on the failures that matter more than their overall correlation suggests. The practical advice is duller than we expected to give: use the judge your access allows and spend the labelling budget on its threshold.

Negative results worth stating. We do not beat the attention-only state of the art; at matched protocol the two are within seed noise, and our contribution to that tier is an explanation and a compression rather than an improvement. The cross-layer dynamics features that our own analysis motivates help in some runs and not in others, so we report them as an available channel rather than a default. And the first version of this study, which contained only one model family, produced two conclusions that more families overturned: that small models cannot abstain when no tool applies, and that output confidence is uniformly useless. Both were artefacts of the sample.

Plumbing dominates method. The largest effects we met while building this study were not differences between detectors but defects in the measurement chain, and each produced a plausible and wrong number: a chat template that silently discarded the tool schemas, so a model appeared to fail on every example; a reasoning model left in its default mode, which spent its generation budget before calling anything and turned a length feature into an apparent detector; a labelling rule that counted a model's use of its own current date as a fabrication; a tokeniser whose structural call tags were deleted by a routine decoding option; and a truncation flag that was wrong for an entire model family. Because internal-state detection has no external check on any of these, the confound panel and the stratification of labels by failure mode were not decoration; they were the only reason we noticed. Appendix A records each defect with the guard that now prevents it.

Limitations. The models run from 0.8B to 4B parameters, chosen so that a dozen runs with per-head eigendecompositions fit a single consumer GPU, and nothing here establishes how the family split behaves at larger scale, which is the first experiment we would add. Detectors are trained per model, since per-head features are indexed by layer and head and do not transfer across architectures; the practical recipe is a small labelled calibration set for each deployed model. The multi-turn evaluation is teacher-forced: the history holds the benchmark's calls, or one deliberately wrong call reported as successful, rather than the model's own trajectory with executed tool outputs. It shows that a judge is not disturbed by a wrong call in the history; it cannot show what a wrong tool *result* does, which needs the benchmark's simulated back ends and is the next experiment. Labels come from comparing a call against a benchmark's ground truth, which is a proxy for whether the action would have been harmful: a wrong but harmless call and a correct but unsafe one are not distinguished, and on the single-turn runs arguments the schema does not define are not penalised (Appendix A). The family split is confounded with the call format the families emit, which a forced-format control would resolve. Every attention feature here is computed on a symmetrised graph and so cannot see the direction of attention flow (Dahlem et al., 2026); the antisymmetric part of the attention matrix is an unread channel, as are the attention-sink statistics of Binkowski et al. (2026). Finally, the attention tier is the most expensive to read (Appendix F); on the hardware we measured the cost is a few percent of generation, and we have not measured it on a production stack.

7 Conclusion

A hallucinated tool call can be detected from a single forward pass, and this study is mostly about which signal to read and when. The first thing to establish is whether an internal judge is needed at all, because that depends on the model: across the families we tested, either output confidence is uninformative and a trained probe is the only thing that works, or confidence is already as good as anything we trained, and the two groups separate cleanly without being explained by size, attention design, release date, or the number of labelled failures available. Where internals are needed, a probe on the residual stream at the structural positions of the call is the strongest judge, and attention weights alone come within a few points of it, so an operator without access to activations still has a usable option.

For attention, one design choice decides everything. Averaging over heads before building the graph cannot represent the contrast between heads, and for algebraic connectivity it can only overstate how well connected the graph is, with equality exactly when all heads share a Fiedler vector, a condition every layer we measured sits far from. Keeping the heads apart moves the attention family from below a length baseline to within reach of the residual stream, and controls that hold the feature count fixed show that it is the resolution that does the work. The strongest published attention-only detector already keeps heads apart, and its features are an in-degree sequence rather than a spectrum, which is why two very different readouts perform alike once resolution is preserved. The same pattern holds in the residual tier, where the detectors that read designated positions beat those that pool over the span. Resolution, not the choice of statistic, is what carries the signal.

Reproducibility statement

Every number in this paper expands from a macro that a released script builds from the result files; no value is typed into the text by hand, a macro with no backing result renders as a visible marker, and an undefined one stops the build. The models and all benchmarks are public. One command reproduces each run and one further command regenerates every table. Labels are recomputed from the stored generations at evaluation time, so a change to the labelling rule cannot leave stale labels behind. Splits are grouped by ground-truth tool, so every reported score is on tools absent from training, and each is a single pooled cross-fit score over the whole dataset rather than an average over small folds. Appendix A lists the validity defects we found in our own measurement chain, each with the artefact it produced and the guard that now prevents it.

Ethics statement

The study measures open-weight language models on public tool-calling benchmarks. No model was trained or fine-tuned, no capability is elicited, and no human subjects or personal data are involved. The work is defensive: its aim is to let an operator stop a wrong tool call before it executes. The same detectors could in principle guide the construction of calls that evade them, which is why we report operating points, the architectures on which each detector weakens, and the model family that never abstains when no tool applies, rather than a single headline number.

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A VALIDITY DEFECTS FOUND IN OUR OWN PIPELINE

Every defect below was present in a working pipeline, produced a plausible number, and was found only because a confound baseline or a failure-mode count disagreed with it. We list them because the class is systematic in internal-state detection work, where nothing external checks the measurement chain.

1. **The chat template silently dropped the tools.** One model family's template ignores the argument that carries tool schemas. The model never saw a tool, answered in prose, and all 678 extracted examples were labelled `no_call`: a hallucination rate of 100% that was a property of prompt rendering. *Guard:* the renderer asserts that every tool name appears in the rendered prompt, falls back to an explicit system-prompt specification when the native template drops them, and refuses to emit an example otherwise.
2. **A reasoning model spent its budget before calling.** A model whose template defaults to chain-of-thought emitted long reasoning before any call, so within a fixed generation budget the call was often never reached. The run produced mostly format failures, and a classifier on generation length alone scored 0.592. It measured reasoning length. *Guard:* disable the thinking mode where the template supports it, and treat any run whose confound floor approaches its detectors as uninterpretable.
3. **The labeller punished the model for knowing the date.** Ground-truth calls in a corpus written in an earlier year contain arguments such as `current_date`. A model using its own sense of today was labelled hallucinating. *Guard:* arguments whose ground-truth value encodes the corpus author's present are excluded from the comparison, and a small alias table handles interchangeable encodings such as a language name against its ISO code.
4. **Hidden states entered the "attention-only" features.** In the library that prior spectral work was built on, four of five nominally spectral metrics project residual-stream states onto the attention eigenbasis, and require both tensors. Reported as attention-only, they inherit the performance of a residual-stream probe. *Guard:* every metric is tagged with the tensors it consumes, and the attention tier admits only metrics that are functions of the attention weights alone (Appendix B).
5. **Probe checkpoints collided.** The probe trainer wrote its early-stopping checkpoint to a fixed path in the working directory, so two trainings running at once overwrote each other. With mismatched dimensions this crashes; with matching dimensions it silently reports one probe's weights as another's. *Guard:* the best state is held in memory.
6. **Repeated prompts were extracted twice.** The set of already-seen prompts was loaded once and never updated, so a benchmark that repeats prompts produced duplicate rows, 13% of each Glaive dump. The duplicates are identical and group with their originals, so they never crossed a train/test boundary, but they up-weighted the repeated examples. *Guard:* deduplicate at write time and again at load time.
7. **Task identity leaked through prompt-level splits.** Errors concentrate on particular tools, so with splits grouped by prompt a classifier reading only the tool identity reaches 0.323 AUC. *Guard:* group splits by tool, and keep the tool-identity classifier in the confound panel, where it sits at or below chance.
8. **A third call format went unparsed.** One model family writes a call as `<function name="f"><param name="k">v</param></function>`, with CDATA around multi-line values, rather than in either format our parser knew. Every call from that family was scored as absent. *Guard:* the parser covers all three dialects and the label distribution of a new model is inspected before its numbers are used.
9. **Decoding deleted the structural tags.** The same family tokenises `<function` and `<param` as special tokens, so decoding the generation with `skip_special_tokens` removed them and left a fragment that could not be parsed. The model appeared to never call a tool. *Guard:* generations are decoded with special tokens kept and only chat control markers removed.
10. **The truncation flag was wrong for a whole family.** Truncation was detected by the absence of the end-of-sequence id, but Llama closes a tool call with a different marker, so the flag read "truncated" for 97.8% of generations on one split and carried no information in the confound panel. *Guard:* truncation is the generated length against the budget, which no tokeniser convention can break.

11. **Our own budget was charged to the model.** A call that fails to parse because the generation budget cut it off is an artefact of the protocol, not a failure of the model. *Guard:* such cases carry a distinct mode and are excluded from the semantic subset.
12. **A massive activation overflowed the storage type.** Per-token residual states were stored at half precision, but one dimension in two thousand carries an activation whose magnitude exceeds that range, and it was written as infinity, which crashed the probe on most records of one run. *Guard:* single precision for stored states, and a finiteness check at load.
13. **Confidence was summarised one way.** The comparison against output confidence used the mean token log-probability, which understates it: the mean of the five least likely tokens scores ten points higher on the family where confidence loses. *Guard:* nine summaries are recorded and each is scored as its own detector, so the trained judges are compared against the best available summary rather than a convenient one (Section 5.11).
14. **Arguments the schema does not define were not penalised.** A call whose expected arguments are right but which carries an invented or schema-echoed extra field was scored valid, whereas the benchmark's own checker rejects it. *Guard:* the labeller has a rule that counts such calls as a distinct failure mode; applied to every stored single-turn run it changes at most 2.8% of labels (gemma3 glaive) and none on most BFCL runs, so the single-turn tables report the lenient labels and the strict ones are available with one switch. On the multi-turn runs a weak model echoes tool schemas into argument-less calls often enough that the rule matters, and Section 5.9 uses it.

Most of these inflate or fabricate a result rather than deflating one. The general lesson is that a detection number in this area should not be believed without a confound panel evaluated under the same protocol, and a stratification of the labels by failure mode.

B Feature provenance

Signals are grouped by the tensors they consume, because the grouping determines both what a deployment needs to expose and what a claim may say.

Family	Tensors	Members
Attention	attention weights	per-head profile, cross-layer dynamics, trajectory and velocity summaries, eigenvalue profile, LapEigvals
Residual	block outputs	token-role probe, Gram spectrum
Hybrid	both	Dirichlet energy, smoothness, spectral entropy and HFER with residual states as the graph signal
Logits	output logits	mean token log-probability

The hybrid family is excluded from the attention tier of this paper. It is not weak, and it may well be a strong detector, but a result obtained from it does not support a statement about attention topology, and it cannot be computed where only attention maps are exposed. What Hugging Face returns as hidden states are residual-stream snapshots after each block; module-internal activations are a fourth family that nothing in this literature yet probes.

C Protocol details

Generation. Greedy decoding, temperature zero, budget 256 new tokens (160 on the multi-turn runs, whose prompts are long), tool schemas passed through the model's own chat template so the model answers in its native call format. Truncation is recorded and available as a feature to the confound panel.

Multi-turn. The four multi-turn categories of BFCL-v4 (base, missing parameter, missing function, long context), one record per conversation turn up to the fourth, with the ground-truth calls of the two preceding turns rendered as assistant messages each followed by a fixed acknowledgement that the calls succeeded. Conversations involving more than 22 tools are skipped rather than truncated. In the cascade condition one of the two preceding turns carries a call drawn from another conversation instead, with the same acknowledgement. Attention is reduced to its features inside a forward hook on each attention module, so that a layer's probabilities are released before the next layer computes; the full-sequence graph features, cubic in the prompt length, are not computed on these runs. Labels use the strict rule that penalises arguments the schema does not define.

Layers and heads. Residual probes read eight depths spaced by `linspace` over the model's layer count, so depth is comparable across models of different size. Per-head features use every attentive layer, which for hybrid stacks is the subset carrying softmax attention.

Probes. Logistic regression with ℓ_2 regularisation, class-balanced, C chosen on validation from $\{0.01, 0.1, 1, 10\}$; the token-role network is a single hidden layer of 512 units with dropout 0.1, AdamW, cosine schedule with warm restarts, early stopping with patience five. Gradient-boosted trees with an automatic fallback to logistic regression are used for the per-layer feature sets, with the fallback triggered by feature count or by a train-validation gap, decided on those folds only.

Scoring. Five-fold cross-fit with folds grouped by ground-truth tool, one pooled AUC per seed over all examples, five seeds, reported as mean and standard deviation across seeds. Bootstrap comparisons of two detectors resample positives and negatives separately and are computed on the pooled scores.

D Seed spread of Table 2

Every entry of Table 2 is a mean over five split seeds of a pooled cross-fit AUC. The spread across seeds, given here, is a spread over ways of partitioning the same tools into folds and not a sampling uncertainty over examples; it is nevertheless the right scale on which to read differences between detectors in the same row, and it is why the main text ranks tiers and not detectors within a tier.

Model	Data	logprob	surface	avg. spectral	per-head	LapEigvals	Lookback	token-role	token-level
Llama-3.2-1B	Glaive	0.576 ± 0.005	0.556 ± 0.051	0.535 ± 0.057	0.940 ± 0.020	0.942 ± 0.010	0.917 ± 0.049	0.957 ± 0.027	0.963 ± 0.013
Llama-3.2-3B	Glaive	0.738 ± 0.000	0.609 ± 0.040	0.652 ± 0.039	0.930 ± 0.009	0.907 ± 0.031	0.911 ± 0.034	0.937 ± 0.027	0.897 ± 0.032
Gemma-3-1B	Glaive	0.647 ± 0.015	0.699 ± 0.033	0.716 ± 0.055	0.856 ± 0.048	0.854 ± 0.034	0.765 ± 0.028	0.915 ± 0.013	0.914 ± 0.023
Llama-3.2-1B	BFCL	0.623 ± 0.000	0.754 ± 0.009	0.773 ± 0.005	0.873 ± 0.014	0.873 ± 0.012	0.861 ± 0.012	0.913 ± 0.009	0.903 ± 0.008
Llama-3.2-3B	BFCL	0.692 ± 0.000	0.632 ± 0.011	0.757 ± 0.011	0.910 ± 0.013	0.895 ± 0.012	0.892 ± 0.010	0.929 ± 0.007	0.930 ± 0.006
Qwen3-1.7B	BFCL	0.537 ± 0.047	0.599 ± 0.003	0.795 ± 0.014	0.892 ± 0.012	0.905 ± 0.011	0.911 ± 0.013	0.927 ± 0.011	0.908 ± 0.007
MiniCPM5-2B	BFCL	0.805 ± 0.000	0.546 ± 0.014	0.635 ± 0.036	0.730 ± 0.011	0.722 ± 0.042	0.771 ± 0.026	0.788 ± 0.015	0.724 ± 0.020
Qwen3.5-0.8B	BFCL	0.769 ± 0.000	0.616 ± 0.017	0.599 ± 0.017	0.657 ± 0.022	0.627 ± 0.019	0.634 ± 0.012	0.708 ± 0.015	0.674 ± 0.013
Qwen3.5-4B	BFCL	0.821 ± 0.000	0.637 ± 0.015	0.620 ± 0.018	0.665 ± 0.027	0.726 ± 0.030	0.692 ± 0.022	0.749 ± 0.020	0.737 ± 0.030
Qwen3.5-2B†	Glaive	0.784 ± 0.000	0.721 ± 0.024	0.708 ± 0.100	0.638 ± 0.051	0.678 ± 0.059	–	0.746 ± 0.116	–
Llama-3.2-1B	BFCL-live	0.403 ± 0.040	0.709 ± 0.015	0.658 ± 0.029	0.816 ± 0.014	0.782 ± 0.026	0.800 ± 0.018	0.808 ± 0.012	0.808 ± 0.012
MiniCPM5-2B	BFCL-live	0.776 ± 0.000	0.582 ± 0.015	0.582 ± 0.027	0.725 ± 0.016	0.661 ± 0.032	0.739 ± 0.034	0.726 ± 0.011	0.741 ± 0.029

E Every run on one population

Table 2 scores each run on the hardest population it supports, which mixes the call-expected population of the BFCL runs with the semantic population of the Glaive runs. The same detectors scored on the semantic population for every run, so that all rows share one definition, are given here; the ordering of tiers is unchanged.

Model	Data	logprob	surface	avg. spectral	per-head	LapEigvals	Lookback	token-role	token-level
Llama-3.2-1B	Glaive	0.576	0.556	0.535	0.940	0.942	0.917	0.957	0.963
Llama-3.2-3B	Glaive	0.738	0.609	0.652	0.930	0.907	0.911	0.937	0.897
Gemma-3-1B	Glaive	0.647	0.699	0.716	0.856	0.854	0.765	0.915	0.914
Llama-3.2-1B	BFCL	0.700	0.765	0.801	0.888	0.888	0.882	0.918	0.916
Llama-3.2-3B	BFCL	0.803	0.610	0.804	0.935	0.930	0.926	0.950	0.952
Qwen3-1.7B	BFCL	0.506	0.591	0.793	0.882	0.894	0.907	0.917	0.891
MiniCPM5-2B	BFCL	0.672	0.543	0.653	0.751	0.725	0.791	0.805	0.754
Qwen3.5-0.8B	BFCL	0.602	0.483	0.512	0.592	0.566	0.555	0.660	0.649
Qwen3.5-4B	BFCL	0.680	0.553	0.620	0.670	0.729	0.709	0.726	0.737
Qwen3.5-2B†	Glaive	0.784	0.721	0.708	0.638	0.678	–	0.746	–
Llama-3.2-1B	BFCL-live	0.392	0.579	0.604	0.755	0.736	0.748	0.776	0.774
MiniCPM5-2B	BFCL-live	0.593	0.537	0.484	0.660	0.574	0.666	0.665	0.681

F What a judge costs at generation time

A guardrail that judges the call before it executes adds its cost to every call. On Llama-3.2-1B and a GeForce RTX 5080 Laptop GPU, over 20 BFCL prompts of median 238 tokens producing calls of median 29 tokens, generation itself takes 654 ms. Every internal judge then needs one teacher-forced pass over prompt and call with the eager attention path, which costs 26 ms; a serving stack that keeps the generation pass's hidden states can skip it for the residual tier, but not for attention, whose probabilities a fused kernel never materialises. On top of that pass, gathering the token-role probe's features takes under a millisecond, the LapEigvals column sums 2 ms, and the per-head profile's $L \times H$ eigendecompositions on the call span 36 ms, or 6% of the generation time, when reduced layer by layer inside the forward hook (71 ms for pass and features together). The eigendecompositions are issued one layer at a time; batching them across layers would remove most of that cost. The probe itself is a logistic regression and is free at this scale. The attention tier is therefore the most expensive to read, by an amount that is small next to generation on this hardware and that grows with the square of the call length, not of the prompt.

G Head-disagreement measurement

Proposition 2 is verified on 210 layers from 3 models and three inputs spanning a well-formed call, a second call and ordinary prose. For the combinatorial Laplacian, where the inequality is a theorem, it holds with 0 violations and a gap bounded below by 0.0026, mean 0.057. For the symmetric normalised Laplacian, where the Rayleigh quotient is not linear in the adjacency and the inequality is therefore not implied, it holds on all but 3 layers, with mean gap 0.098 and a median ratio of 1.14 between the averaged graph's connectivity and the mean over heads, reaching 1.72. The mean pairwise absolute alignment of per-head Fiedler vectors is 0.45 with maximum 0.94, so the equality condition of the proposition is never met, and the across-head standard deviation of the per-head Fiedler value averages 0.154.